Problem Solving Handout

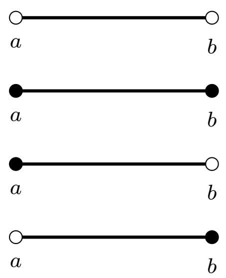
CHAPTER 1

Intervals of Real Numbers

Chapter Road-Map (What You Will be Able to Do)

- Read any interval notation and translate it into inequalities (and vice versa).
- Interpret open/closed/half-open intervals geometrically on the real line.
- Solve inequalities and express answers cleanly using interval notation.
- Rewrite absolute-value statements as intervals and unions of intervals.
- Use ε-neighborhoods: $|x-a|<\varepsilon \iff (a-\varepsilon, a+\varepsilon)$.
- Perform basic interval arithmetic (addition, scaling) and avoid common mistakes.

Interval



Meaning (Endpoint Rule)

(a,b) : all x with $a<x<b$ (both endpoints excluded).

$[a,b]$: all x with $a\le x\le b$ (both endpoints included).

$[a,b)$: all x with $a\le x<b$ (include left, exclude right).

$(a,b]$: all x with $a<x\le b$ (exclude left, include right).

Legend: filled dot = included endpoint, empty dot = excluded endpoint. The thick segment indicates "all points in between."

Contents

1 Intervals from Inequalities (Core Definitions)

We begin strictly with the interval definitions from the given context. An interval is a set of real numbers described using inequalities. The brackets encode whether endpoints are included.

Definition

Let $a, b \in \mathbb{R}$ with $a < b$. Then:

$$(a,b) = \{x \in \mathbb{R} : a < x < b\}, \qquad [a,b] = \{x \in \mathbb{R} : a \leq x \leq b\},$$
$$[a,b) = \{x \in \mathbb{R} : a \leq x < b\}, \qquad (a,b] = \{x \in \mathbb{R} : a < x \leq b\},$$
$$(a,\infty) = \{x \in \mathbb{R} : x > a\}, \qquad [a,\infty) = \{x \in \mathbb{R} : x \geq a\},$$
$$(-\infty,b) = \{x \in \mathbb{R} : x < b\}, \qquad (-\infty,b] = \{x \in \mathbb{R} : x \leq b\}.$$

Interpretation

How to read each symbol.

- (a,b): between a and b, but not equal to either endpoint.
- $[a,b]$: between a and b, including both endpoints.
- $[a,b)$: include a, exclude b.
- $(a,b]$: exclude a, include b.
- (a,∞): means strictly greater than a.
- $[a,\infty)$: means greater than or equal to a.
- $(-\infty,b]$: means lesser than or equal to b.
- $(-\infty,b)$: means strictly lesser than to b.

Two-Way Translation Table

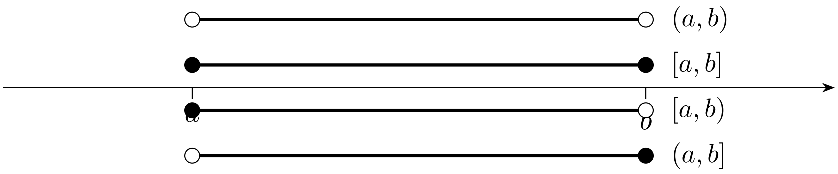
Translation Table

Inequality form	Interval form
$a < x < b$	(a,b)
$a \leq x \leq b$	$[a,b]$
$a \leq x < b$	$[a,b)$
$a < x \leq b$	$(a,b]$
$x > a$	(a,∞)
$x \geq a$	$[a,\infty)$
$x < b$	$(-\infty,b)$
$x \leq b$	$(-\infty,b]$

2 Visual Dictionary And Diagram Interpretation

A picture on the number line is not decoration: it is a *second definition*. When you draw the set, you can instantly check endpoint inclusion and avoid logical mistakes.

2.1 Open/Closed/Half-Open Intervals

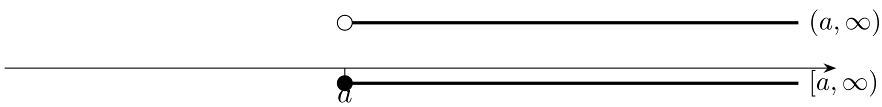


Interpretation

Diagram interpretation:

- The thick line segment represents *all* points between a and b.
- The endpoints are separate logical decisions:
  - filled dot $\Rightarrow$ endpoint is in the set,
  - empty dot $\Rightarrow$ endpoint is not in the set.
- If you can read the dots correctly, you can translate to inequalities instantly.

2.2 Infinite Intervals



Interpretation

Diagram Interpretation: The arrow means "continues forever." The only inclusion decision is whether a is included.

3 Intervals as solution sets of inequalities

Worked Example

Solve $2x - 3 < 5$.
Step 1 (algebra): $2x < 8 \Rightarrow x < 4$.
Step 2 (interval form): $(-\infty, 4)$.

Worked Example

Solve $-1 \leq x < 4$.
Interpretation: include -1 (because $\leq$), exclude 4 (because $<$).
Answer: $[-1, 4)$.

Worked Example

Solve $x(x-1) \geq 0$.
Critical points: $x = 0$ and $x = 1$.
Sign logic: a product is nonnegative when factors have the same sign or one is zero.
Hence the solution set is $(-\infty, 0] \cup [1, \infty)$.

4 Absolute value as Distance (Interval Viewpoint)

Theorem / Fact

For any $a \in \mathbb{R}$ and $\varepsilon > 0$,

$$|x-a| < \varepsilon \quad \iff \quad a-\varepsilon < x < a+\varepsilon \quad \iff \quad x \in (a-\varepsilon, a+\varepsilon).$$

Worked Example

Solve $|x| < 2$.
This means the distance from 0 is less than 2:

$$|x| < 2 \iff -2 < x < 2 \iff x \in (-2, 2).$$

Worked Example

Solve $|x-1| \geq 3$.
Distance from 1 is at least 3, so x is either far left or far right:

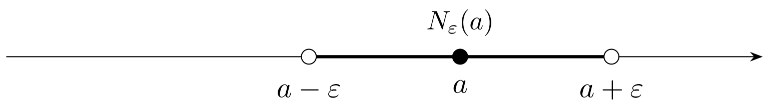
$$|x-1| \geq 3 \iff x \leq -2 \text{ or } x \geq 4 \iff (-\infty, -2] \cup [4, \infty).$$

5 ε-neighborhoods (Open Intervals Centered at a Point)

Definition

Let $a \in \mathbb{R}$ and $\varepsilon > 0$. The ε-neighborhood of a is

$$N_\varepsilon(a) = \{x \in \mathbb{R} : \ |x-a| < \varepsilon\} = (a-\varepsilon, a+\varepsilon).$$



Interpretation

Why this is important: In analysis, to say "something is close to a" means it lies in $N_\varepsilon(a)$ for a small ε. By shrinking ε, we zoom in on a. This is the language behind limits and continuity.

6 Interval Arithmetic (Basic Operations)

Theorem / Fact

If $I = [a, b]$ and $J = [c, d]$, then

$$I + J = \{x + y : \ x \in I, \ y \in J\} = [a + c, \ b + d].$$

Worked Example

Let $I = [1, 2]$ and $J = [3, 5]$. Then $I + J = [4, 7]$.

7 Exercises and solution sketches

Exercises

1. Express the solution set of $2x - 5 \geq 1$ as an interval.
2. Draw $(-3, 2]$ and write one sentence interpreting it.
3. Find $(-1, 4) \cap [2, 6]$.
4. Domain of $f(x) = \ln(x + 2)$.
5. Solve $|x| \leq 5$.
6. Is (a, b) open or closed?
7. Explain why $(a, \infty]$ is meaningless.
8. Write $\mathbb{R} \setminus [0, 1]$ as a union of intervals.
9. Solve $x(x - 3) > 0$ using intervals.
10. In 2–3 lines: Why are ε-neighborhoods important?

Solution Sketches

1. $2x - 5 \geq 1 \Rightarrow x \geq 3$, so $[3, \infty)$.
2. Open at -3, closed at 2: between them, include 2 but exclude -3.
3. $[2, 4)$.
4. Need $x + 2 > 0 \Rightarrow x > -2$, so $(-2, \infty)$.
5. $|x| \leq 5 \iff -5 \leq x \leq 5$, so $[-5, 5]$.
6. (a, b) is open (endpoints excluded).
7. $\infty \notin \mathbb{R}$, cannot be an included endpoint.
8. $(-\infty, 0) \cup (1, \infty)$.
9. Zeros 0 and 3; product > 0 on $(-\infty, 0) \cup (3, \infty)$.
10. They formalize "closeness" using intervals: $|x - a| < \varepsilon$ equals $(a - \varepsilon, a + \varepsilon)$, used in limits/continuity.

End of CHAPTER 1